

Name of the Student	Narra Sai Mounika
Reg. No	192425367
GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT :A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

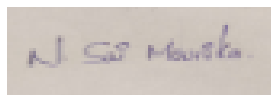
Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I Narra Sai Mounika (192425367) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Narra Sai Mounika

Signature of the student:

Name of the student:

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05**Total Marks:25****Q1. CNN vs Fully Connected Neural Network**

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Answer:**Comparison Table**

Parameter	CNN	Fully Connected Neural Network (FCNN)
Architecture	Uses convolution, pooling, and fully connected layers	Every neuron connected to every neuron in next layer
Motivation	Designed specifically for image processing	Designed for general-purpose data
Spatial Feature Extraction	Preserves spatial relationships between pixels	Loses spatial information
Parameters	Fewer parameters due to parameter sharing	Large number of parameters
Computational Complexity	Lower	Higher
Memory Requirement	Less memory	More memory
Feature Learning	Automatic feature extraction	Requires flattened input
Suitability for Images	Highly suitable	Less suitable

Analysis**CNN**

CNN automatically learns image features such as:

- Edges
- Corners
- Textures
- Shapes

FCNN

FCNN converts images into a 1D vector, causing loss of spatial information.

Role of CNN Components

Convolution Layer

- Extracts local image features.
- Detects edges and patterns.

Filters

- Small matrices that scan images.
- Learn useful visual features automatically.

Pooling Layer

- Reduces feature map size.
- Reduces computation.
- Improves robustness.

Critical Evaluation

CNN Advantages

- Better image understanding.
- Fewer parameters.
- Higher classification accuracy.

CNN Limitations

- Requires GPU resources.
- Longer training time.

FCNN Advantages

- Simple architecture.
- Easy implementation.

FCNN Limitations

- Cannot efficiently process images.
- Large memory consumption.

Conclusion

CNN is more suitable for medical X-ray classification because it preserves spatial information, learns features automatically, and achieves higher accuracy with fewer parameters

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Answer:

Comparison of CNN Layers

Layer	Main Function	Spatial Information	Computation Role	
Convolution Layer	Feature extraction	Preserved	Moderate	Detects patterns
Activation Layer (ReLU)	Introduces non-linearity	Preserved	Low	Improves learning
Pooling Layer	Down-sampling	Partially reduced	Low	Reduces dimensions
Fully Connected Layer	Classification	Not preserved	High	Final decision making

Comparison of Filters

Feature	Early-Layer Filters	Deep-Layer Filters
Learned Features	Edges, lines, corners	Objects, shapes, defects
Complexity	Simple	Complex
Receptive Field	Small	Large
Information	Local	Global

Analysis

Early Layers

Learn:

- Horizontal edges
- Vertical edges
- Texture patterns

Deep Layers

Learn:

- Cracks
- Surface defects
- Product abnormalities

Critical Evaluation

Strengths

- Hierarchical feature learning.
- Accurate defect identification.

Weaknesses

- Requires large datasets.
- Computationally intensive.

Conclusion

The combination of convolution, activation, pooling, and fully connected layers enables accurate defect detection by progressively learning low-level and high-level features.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Answer:

Comparison Table

Parameter	Parameter Sharing (CNN)	Independent Weights (FCNN)
Parameters	Fewer	Very Large
Memory Usage	Low	High
Computation	Faster	Slower
Feature Detection	Efficient	Less Efficient
Translation Invariance	Supported	Not Supported
Scalability	High	Low

Analysis

Independent Weights

For an image of 224×224 pixels:

$$224 \times 224 = 50,176 \text{ inputs}$$

Connected to 100 neurons:

$$\text{Parameters} = 50,176 \times 100$$

$$= 5,017,600 \text{ parameters}$$

Parameter Sharing

Using 32 filters of size 3×3 :

Parameters =

$$(3 \times 3 \times 1 + 1) \times 32$$

$$= 320 \text{ parameters}$$

Benefits

- Reduced memory usage
- Faster training
- Better generalization

Critical Evaluation

CNN Strengths

- Reuses filters throughout image.
- Detects objects regardless of location.

FCNN Weaknesses

- Requires enormous parameters.
- High computational cost.

Conclusion

Parameter sharing is essential for large-scale image processing because it significantly reduces complexity while maintaining high feature-detection capability.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Answer:

Comparison Table

Parameter	AlexNet	ResNet
Year	2012	2015
Depth	8 Layers	50, 101, 152+ Layers
Architecture	Sequential CNN	Residual CNN
Filters	Standard Convolution	Residual Blocks
Parameters	Higher	More Efficient
Feature Learning	Good	Excellent
Vanishing Gradient	Possible	Solved
Training Difficulty	Moderate	Easier for Deep Networks
Computational Cost	Lower	Higher
Accuracy	High	Very High

Analysis

AlexNet

Strengths

- Simpler architecture
- Faster training
- Lower computational cost

Weaknesses

- Limited depth
- Lower feature-learning capability

ResNet

Strengths

- Deep architecture
- Skip connections
- Better feature extraction
- Higher accuracy

Weaknesses

- More computational resources needed

Residual Learning

Output:

$$F(x) + x$$

where:

- $F(x)$ = learned mapping
- x = shortcut connection

This prevents vanishing gradients.

Recommendation

Recommended Architecture: ResNet

Justification

- Learns complex visual patterns.
- Higher accuracy.
- Better feature extraction.
- Suitable for large-scale image recognition.

Conclusion

For complex image-recognition applications, ResNet is the preferred architecture because of its superior performance and deep feature-learning ability.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Answer:

Comparison Table

Parameter	Without Regularization	With Regularization
Overfitting	High	Low
Generalization	Poor	Good

Parameter	Without Regularization	With Regularization
Training Stability	Less Stable	More Stable
Validation Accuracy	Lower	Higher
Real-world Performance	Weak	Strong
Robustness	Low	High

Analysis of Regularization Techniques

1. Dropout

Working

Randomly deactivates neurons during training.

Example:

- Dropout = 0.5

Benefits

- Prevents co-adaptation.
- Reduces overfitting.

2. Data Augmentation

Techniques:

- Rotation
- Flipping
- Zooming
- Cropping
- Brightness adjustment

Benefits

- Increases dataset diversity.
- Improves robustness.

3. Batch Normalization

Functions

- Normalizes activations.
- Stabilizes training.
- Speeds convergence.

Benefits

- Faster learning.
- Better accuracy.

Critical Evaluation

Non-Regularized CNN

Strengths

- Fast training.

Weaknesses

- Memorizes training data.
- Poor performance on unseen images.

Regularized CNN

Strengths

- Better generalization.
- Reduced overfitting.
- Improved reliability.

Weaknesses

- Slight increase in training time.

Recommended Approach

Use a combination of:

1. Dropout (0.5)
2. Data Augmentation
3. Batch Normalization

This combination provides the best balance between accuracy, robustness, and generalization.

Conclusion

A regularized CNN is the most suitable approach for plant disease classification because it minimizes overfitting and achieves reliable performance on real-world agricultural images.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand